

Fairness Experiment Report - Upload 49

Fairness Experiment Report - Upload 49 summarizes the current fairness state for upload 49. Accuracy changed by -0.016 and fairness score changed by +0.097. Good fairness gain with acceptable accuracy trade-off. The best balanced model is ThresholdOptimizer_mitigated_threshold with combined score 0.908.

Upload ID: 49

Dataset: adult.csv

Model: adult_income_pipeline.pkl

Best Balanced Model: ThresholdOptimizer_mitigated_threshold

Best Accuracy Model: Pipeline_mitigated_reweighting

Best Fairness Model: ThresholdOptimizer_mitigated_threshold

Accuracy: 0.838

Fairness Score: 0.954

DPD: 0.088

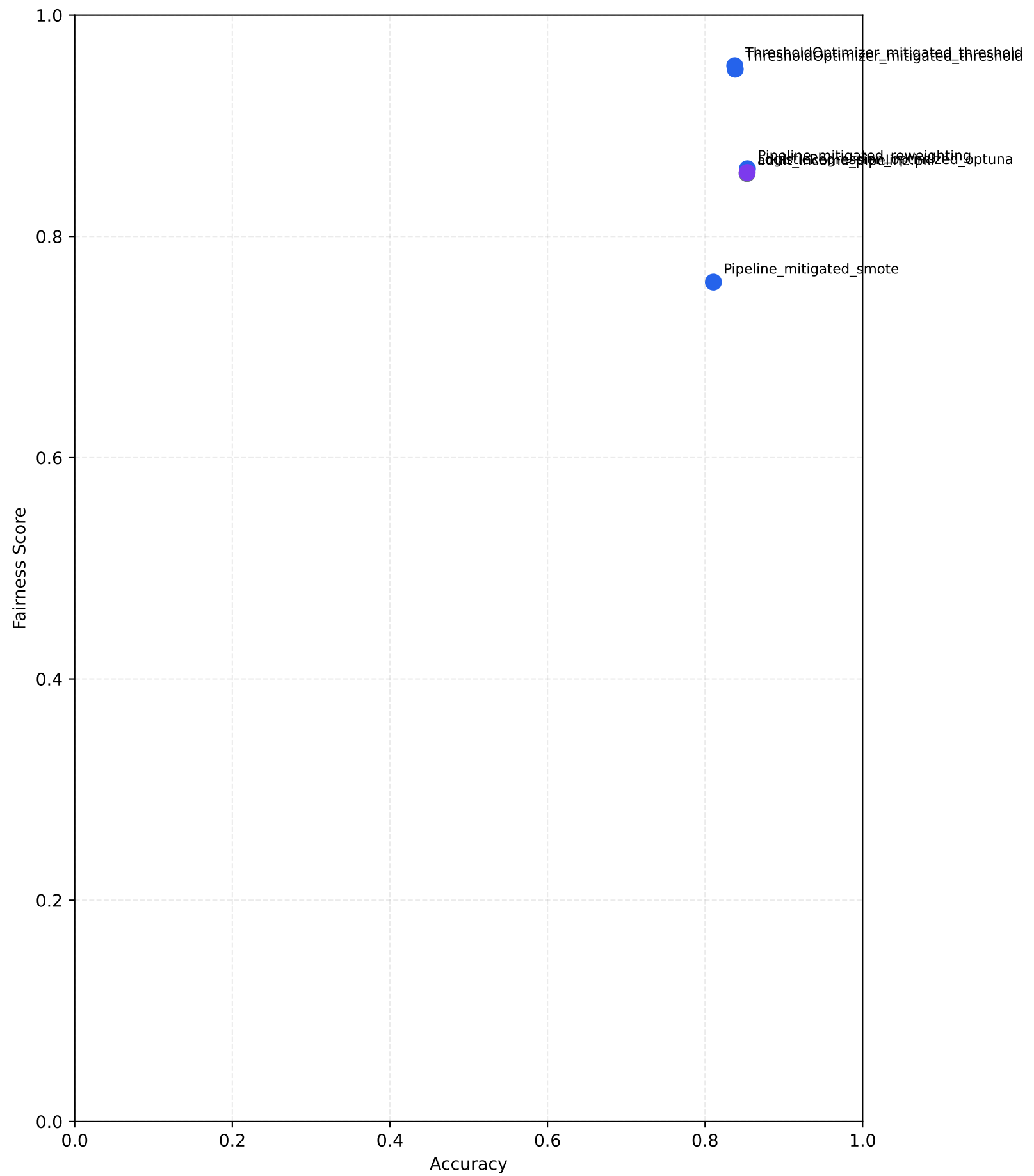
EOD: 0.003

DIR: 0.553

Sections: comparison, optimization

Generated at: 2026-05-13T05:53:55.110789Z

Accuracy vs Fairness Trade-off



Model Ranking and Interpretation

Good fairness gain with acceptable accuracy trade-off. The best balanced model is ThresholdOptimizer_mitigated_threshold with combined score 0.908.

Mitigated: ThresholdOptimizer_mitigated_threshold | accuracy=0.838, fairness=0.954, combined=0.908

Mitigated: ThresholdOptimizer_mitigated_threshold | accuracy=0.838, fairness=0.951, combined=0.906

Mitigated: Pipeline_mitigated_reweighting | accuracy=0.854, fairness=0.861, combined=0.858

Original: adult_income_pipeline.pkl | accuracy=0.853, fairness=0.857, combined=0.855

Optimized: LogisticRegression_optimized_optuna | accuracy=0.853, fairness=0.858, combined=0.855

Mitigated: Pipeline_mitigated_smote | accuracy=0.811, fairness=0.759, combined=0.779